

ABDUL RAFAY

AI ENGINEER

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PROFILE

Accomplished Senior AI Engineer with over 3 years of hands-on expertise building, optimizing, and deploying production-grade AI systems and SaaS platforms serving over 10,000+ concurrent users. Deeply specialized in Large Language Models (LLMs), scalable Retrieval-Augmented Generation (RAG) frameworks, automation pipelines, and real-time multimodal architectures. Proven track record in orchestrating microservices backends with Python, Node.js, and TypeScript, translating highly experimental technical prototypes into production environments with continuous integration and AWS deployment strategies. Recognized for bringing founding-team ownership, high-velocity engineering, and cross-functional leadership to modern software solutions.

PROFESSIONAL EXPERIENCE

Deep Query

Lahore, Punjab

Senior AI Engineer

Sep 2025 - Present

- Architect and ship end-to-end AI-powered web ecosystems, utilizing advanced Python frameworks, Next.js, and the MERN stack deployed on enterprise AWS infrastructure.
- Lead and mentor a high-performing engineering squad, overseeing strict code reviews, architectural planning, and integration of complex multi-model AI workflows.

Medaibility (AI Medical Education Platform)

- Engineered the comprehensive core AI stack for a high-traffic medical education SaaS, optimizing database queries and microservices to natively support over 10,000 concurrent students.
- Designed a real-time clinical consultation engine with LiveKit WebRTC and asynchronous AWS Lambda workers, incorporating an intelligent on-call AI medical assistant.
- Implemented an automated AI Scribe that listens to consultation feeds, automatically transcribes interactions, and formats standard clinical documentation with high contextual accuracy.
- Integrated automated QBank generators, spaced-repetition logic, and vision-based image review tools backed by a NestJS microservices mesh and fine-tuned MySQL relational schemas.

ScholarAI (Academic Research & Context Discovery Platform)

- Developed an advanced academic intelligence engine capable of parsing, indexing, and synthesizing complex multi-page research documents to deliver precise literature summaries.
- Designed a vector-embedded conversational system allowing researchers to execute real-time granular document chats against massive corporate and academic knowledge stores.
- Optimized ingestion workflows by writing high-throughput text extraction utilities, reducing token overhead while maintaining accurate citation references.

VoiceFlux (Multilingual Dubbing & Neural Speech Suite)

- Built an end-to-end automated audio/video translation and dubbing pipeline utilizing state-of-the-art Text-to-Speech (TTS) and voice-cloning foundation models.
- Programmed advanced multi-lingual prosody controls, enabling automated translation adjustments to match the original vocal speed, pitch, and emotional cadence.
- Configured cloud-based media transcoding queues that cleanly split, process, and merge audio stems with minimal degradation in video quality.

Student Revelation System (Multi-Tenant Institution Engine)

- Architected a multi-tenant enterprise management platform featuring independent portal layouts tailored for administrators, educators, medical staff, and parents.
- Integrated low-latency interactive messaging layers via WebSocket frameworks (Socket.io) to synchronize cross-portal comms and real-time state broadcasts.
- Automated individualized education plan (IEP) workflows and comprehensive academic tracking mechanisms decrease institutional administrative overhead.

Pebi Technologies

Faisalabad, Punjab

AI Engineer

Apr 2025 - Sep 2025

- Developed and shipped production-ready AI applications onto scalable AWS networks and Vercel edge infrastructure.
- Conducted deep integration passes with OpenAI, Groq, and Hugging Face API infrastructures to scale internal product features.

Detext AI (Multi-Model Content Detection & Humanization Suite)

- Created a sophisticated multi-model linguistic wrapper containing advanced AI-vs-human text detectors and custom natural language humanizer algorithms.
- Engineered a multimodal generation pipeline supporting parallelized text, asset generation, and voice synthesis workflows over optimized API layers.
- Configured secure multi-tier Stripe metered billing gateways, analytics collection layers, and Redis-backed cron jobs to track accurate platform usage statistics.

DropPR.ai (Automated Media Distribution & Content Pipeline)

- Designed a powerful media distribution engine that translates short/long-form video and audio inputs into highly optimized, SEO-ready press releases.
- Built high-throughput distributed background task queues to ingest, transcribe, and restructure media feeds simultaneously for thousands of concurrent users.
- Connected platform APIs with automated distribution grids, enabling direct, synchronized syndication of content across 1,500+ high-authority digital publications.

Okani Travels (Global Multilingual Booking Ecosystem)

- Co-engineered a comprehensive global hospitality booking platform enclosing localized modules for accommodations, flight paths, vehicle rentals, and curated experiences.
- Architected an AI-driven predictive pricing engine and intelligent multi-agent itinerary planner alongside dynamic localized checkout flows using Stripe.
- Assembled lightning-fast real-time administrative panels that handle automated predictive inventory scaling, booking tracking, and instantaneous financial reporting.

Dzone Technologies

Faisalabad, Punjab

AI Engineer

Aug 2023 - Apr 2025

- Owned the full product development lifecycle from database schema blueprinting to server provisioning and initial continuous integration configurations.
- Maintained consistent agile delivery patterns to safely deploy cross-stack web assets with high uptime.

PandaChat (Real-Time Knowledge-Graph & Enterprise RAG Platform)

- Engineered a highly responsive enterprise chat layer featuring automated language translations and a proprietary RAG reasoning engine ("Panda IQ").
- Achieved sub-2-second retrieval and generation speeds across massive message chains using highly optimized localized vector search patterns.
- Authored a structured "chat with your data" pipeline allowing secure, deep tabular queries across unstructured PDFs, Excel spreadsheets, and live targeted websites.
- Shipped native data synchronization integrations connecting the platform with third-party storefront architectures, including Shopify, Magento, and standard ERP databases.

Rewordify (Contextual AI Re-authoring Platform)

- Built an advanced context-aware text regeneration tool empowering users to selectively transform tone, formal structure, and length parameters on the fly.
- Constructed a decoupled architecture separating a specialized high-performance Python AI microservice from an interactive Next.js user dashboard.

GenX Career (Job Aggregator & Smart CV Compatibility Engine)

- Built an automated web collection engine that aggregates listings across primary job platforms, including LinkedIn, Indeed, and relevant industry boards.
- Designed an LLM compatibility scorer that evaluates resumes against targeted job descriptions to output explicit match metrics and keyword optimizations.
- Engineered an interactive, schema-based resume builder that converts raw developer profiles into professionally formatted templates.

SEOHub Analytics Platform (Automated Content Engine)

- Designed a modern AI-based SEO analysis system that systematically analyzes traffic data points and targets potential high-value keywords automatically.
- Programmed a dynamic content engine that auto-generates optimized marketing landing pages on top of reactive, MongoDB-backed real-time data streams.

Smart Transform (Multimodal Digital Asset Transformer)

- Engineered a unified media processing engine consolidating multi-directional workflows: text-to-video, text-to-image, text-to-voice, voice-to-text, and image-to-text.
- Constructed a secure centralized administration command center tracking cross-tool API token usage, system compute cost metrics, and user consumption histories.

SKILLS

AI & Machine Learning: LLM Architecture & Integration (OpenAI, Hugging Face, Groq), Advanced RAG Pipelines, Conversational Agents, Multimodal Generative AI (Text, Image, Audio, Video), AI Content Optimization & Detection, Prompt Engineering, Autonomous AI Workflows, Real-Time Media Streaming (LiveKit WebRTC, WebSockets)

Programming Languages: Python, JavaScript, TypeScript, HTML5, CSS3, SQL

Frameworks & Libraries: Next.js, React.js, Node.js, Nest.js, Express.js, Redux Toolkit, Flask, FastAPI, Tailwind CSS

Databases & Vector Search: MongoDB, MySQL, PostgreSQL, Firebase, Supabase, Redis, Vector Embeddings & Indexing

Cloud, DevOps & Tools: AWS (Lambda, S3, EC2), Vercel, Railway, Git, GitHub Actions, CI/CD Pipelines, Postman, Docker

Core Competencies: Technical Team Leadership, Agile/Scrum Methodologies, Developer Mentorship

EDUCATION

National Textile University	Faisalabad, Punjab
Bachelor of Science in Computer Science — CGPA 3.88/4.00	Graduated: August 2025
• Awarded multiple high-performance merit-based institutional scholarships.	
• Received an official government award/laptop honoring academic excellence and high graduation standings.	

CERTIFICATIONS

ChatGPT Prompt Engineering for Developers — DeepLearning.AI

- Mastered foundational and advanced large language model prompt strategies, context window structuring, and programmatic API content formatting.